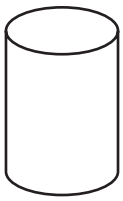
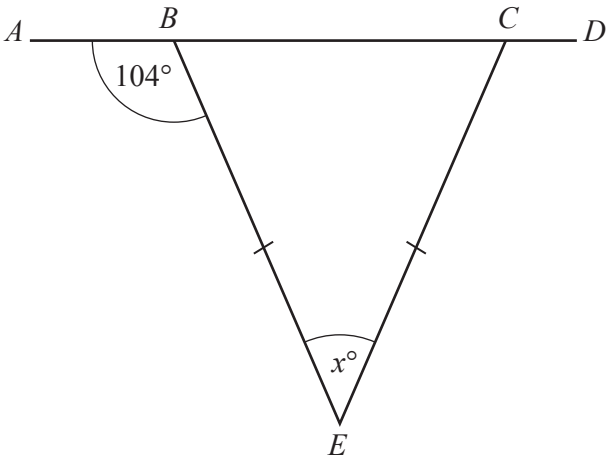
6 **(a)**



Write down the mathematical name of this solid.

..... [1]

(b)



NOT TO SCALE

The diagram shows triangle *BCE* and a straight line *ABCD*.
BE = *CE* and angle *ABE* = 104°.

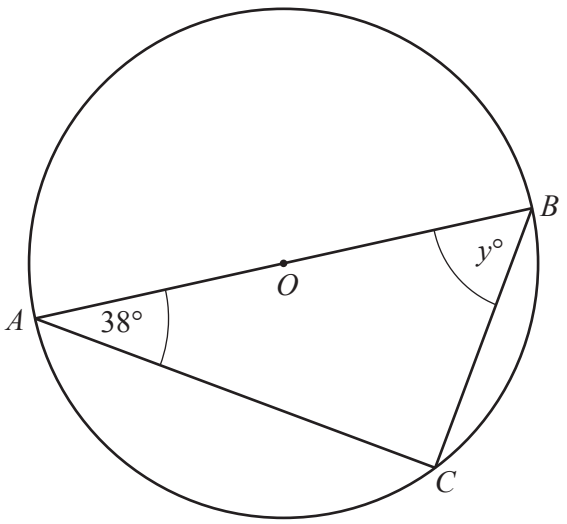
Find the value of *x*.

x = ... [2]

(c) Work out the size of one interior angle of a regular polygon with 15 sides.

..... [2]

(d)



NOT TO SCALE

A, *B* and *C* are points on a circle, centre *O*.

(i) Write down the mathematical name of the line *BC*.

..... [1]

(ii) Draw a tangent to the circle at point *B*. [1]

(iii) The area of the circle is $245.5\,\text{cm}^2$.

Calculate *AB*.

AB = ... cm [3]

(iv) Find the value of *y*.

y = ... [2]